Hitodama-style Voice PC Assistant — Software Requirements Specification

Oct 12, 2026 · @Akhil

1. Introduction

This document specifies **Wisp** (working title), a personal Windows 11 voice assistant that dictates text, answers questions about the screen, runs scheduled routines and controls the PC, all powered by the Gemini API, in enough detail for a coding agent to build version 1 without asking questions.

1.1 Purpose and audience

The reader is an autonomous coding agent. Every requirement has an ID, a MUST/SHOULD/MAY level and, where it matters, an acceptance criterion (AC). Where a fact could not be verified at writing time, the text says so and names the page the agent must read.

1.2 Scope

Version 1 delivers ten features: setup and key storage, global hotkeys, dictation, voice commands, Circle to Ask, scheduler and routines, a Computer Use agent (full PC control), a to-do list, an "island" overlay widget, and customization slots. It is for one user on one machine. Licensing, payments, code signing, accounts, cloud sync and telemetry are out of scope.

1.3 Conventions

- **MUST / SHOULD / MAY** follow RFC 2119.
- Requirement IDs: `FR-<AREA>-<nn>` (functional), `NFR-<nn>` (non-functional). Areas: SET, HK, DIC, CMD, CTA, SCH, CU, TODO, ISL, CUS, GEM.
- `Chord` = keys held together. `Hold` = active while the chord is down, ends on release.
- Code font marks identifiers, paths and commands.

1.4 Rules for the implementing agent

1. Build in the milestone order of section 11. Do not start a milestone until the previous one meets its acceptance criteria.
2. Never invent Gemini request or response shapes. Before each integration, read the official page named in section 5.1 and follow it exactly.
3. Model IDs are configuration values, not constants. Defaults are in section 5.2; verify them against the live model list before use.
4. The only network traffic allowed is to Google Gemini endpoints, plus pages the user asks to open in their own browser. No telemetry, no analytics, no update checks.
5. The API key MUST live only in Windows Credential Manager. It MUST NOT appear in logs, config files, error messages, crash dumps or the frontend bundle.
6. Windows hooks, overlays and input injection cannot be tested in headless CI. For every such feature, ship a manual test checklist in `docs/manual-tests/` alongside unit tests.
7. Keep the code small and boring: no framework beyond those named in section 3.

1.5 Definitions

Term	Meaning
Island	A small always-on-top pill window at top-centre of the screen that shows assistant state and expands into a panel
Soul	The assistant's persona and voice settings (one of four customization slots)
Loadout	The set of tools the assistant is allowed to use (a customization slot)
Command mode	Hold-to-talk voice commands that call local tools
Computer Use	Gemini's screenshot-and-action loop that operates the PC like a person
Routine	A saved natural-language command with a schedule
UIPI	User Interface Privilege Isolation: Windows blocks input from a normal process into an elevated (admin) window

Term	Meaning
LL hook	A low-level keyboard hook (`WH_KEYBOARD_LL`)
GSMTC	Global System Media Transport Controls, the Windows API for media session info and control

2. Product overview

Wisp lets one person run their Windows 11 PC by voice: hold a key, speak, and the assistant types for you, answers about what is on screen, opens apps, runs routines or operates the desktop itself. It runs in the system tray, uses the user's own Gemini API key, and stores everything locally.

2.1 Goals

- Cut typing: dictated text, cleaned of false starts, lands in whatever box has focus within 4 seconds of releasing the key (p90, utterances up to 10 s).
- One assistant for quick actions (open, search, remind) and long tasks (Computer Use).
- Zero running cost on the free Gemini tier for everyday use.
- Safe by default: nothing sensitive happens without an on-screen confirmation.

2.2 User and environment

Item	Value
User	One person, personal use, no accounts
OS	Windows 11 (x64) only. Windows 10 and other OSes are not supported
Runtime	WebView2 (preinstalled on Windows 11), no other runtime to install
Hardware	A working microphone; one or more monitors; internet access
Privilege	Runs as a normal user. Admin rights are never required

Item	Value
Language	English UI and English speech in v1. Gemini may transcribe other languages but they are not tested

2.3 Assumptions

1. The user supplies a Gemini API key from Google AI Studio. The free tier is enough to start; paid tier 1 is optional.
2. Gemini Computer Use is a Preview capability: its behaviour may change, so the integration is isolated behind one module (`gemini::computer_use`).
3. The user accepts that screenshots, audio and clipboard text sent for a request go to Google. On the free tier Google may use that data to improve its products; the first-run screen MUST say so.
4. The app is built on GitHub Actions or on the user's PC. The implementing agent may run on Linux, so it cannot execute the app there.

2.4 Constraints

- Modifier-only chords (Ctrl+Win, Ctrl+Shift) cannot be registered with the normal hotkey API; a low-level hook is required (section 7).
- Windows blocks injected input into elevated windows (UIPI). The app does not try to work around this.
- Gemini free-tier rate limits apply per project and are not published per model in the docs; the user checks them in AI Studio. The app MUST handle HTTP 429 gracefully.
- Several Gemini model prices double on 1 January 2027. This does not affect the free tier.

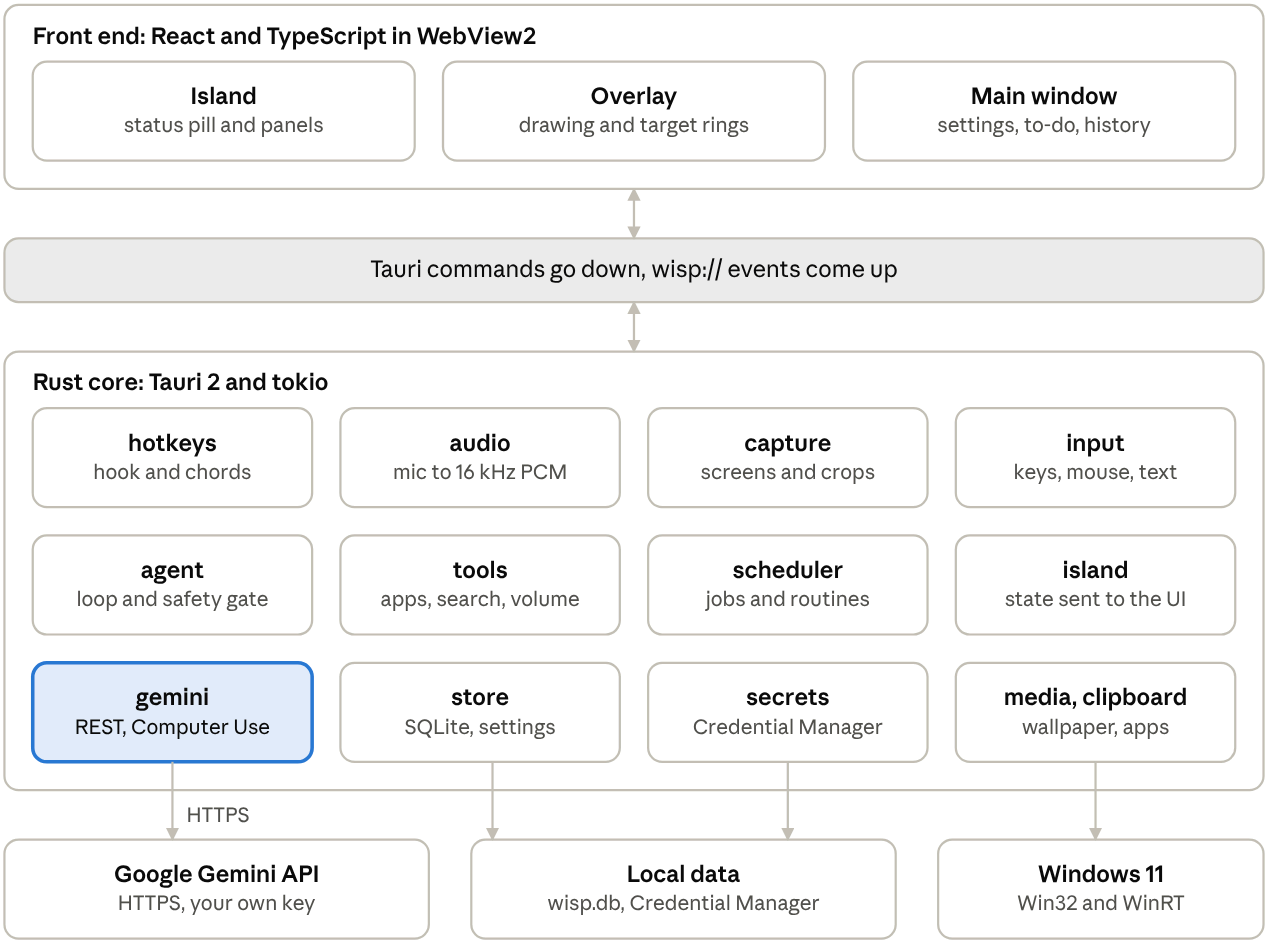
2.5 Out of scope for v1

Licensing, payments, code signing, auto-update, cloud sync, multi-user, mobile, macOS/Linux, a plugin store, wake-word activation, text-to-speech replies from the assistant, running arbitrary shell commands as a tool, and waking a sleeping PC to run a routine.

3. Tech stack and architecture

Wisp is a Tauri 2 app: a Rust core does all system work (hooks, audio, capture, input, scheduling, Gemini calls) and a React front end draws the island, overlays and settings. The front end never holds the API key and never calls Gemini; it only sends commands to Rust and receives events.

Only the Rust core touches the network, the key and Windows



Highlighted: the only module that sends data to Google

architecture · UI, Rust core, external systems

The UI sends commands and receives events. Everything that listens, captures, injects input or calls Google lives in Rust.

3.1 Stack

Layer	Choice	Notes
App framework	Tauri 2.x	Rust stable, MSVC toolchain, WebView2
Core language	Rust (edition 2021)	Async runtime: `tokio`
Front end	React 18 + TypeScript + Vite	Styling: Tailwind CSS. State: Zustand. No router; windows are separate entry points
Storage	SQLite via `rusqlite` (bundled)	One file in `%APPDATA%/Wisp/wisp.db`
Secrets	Windows Credential Manager via the `keyring` crate	Target name `Wisp/GeminiApiKey`
Installer	NSIS, per-user install, unsigned	No admin prompt
CI	GitHub Actions, `windows-latest`, `tauri-action`	Builds the installer; runs unit tests

3.2 Rust crates

Crate	Used for
`windows` (windows-rs)	LL keyboard hook, `SendInput`, `ShellExecuteW`, clipboard listener, UI Automation, wallpaper, GSMTC media session, DPI
`cpal` + `rubato`	Microphone capture and resampling to 16 kHz mono 16-bit PCM
`xcap`	Screen capture per monitor (BGRA to PNG/JPEG via `image`)
`reqwest` (rustls)	Gemini REST calls
`tokio-tungstenite`	Gemini Live API WebSocket (optional streaming path, section 5.6)
`serde`, `serde_json`	Messages and config

Crate	Used for
`rusqlite`, `refinery`	Database and migrations
`keyring`	Credential Manager
`croner`	Cron-style schedule evaluation
`chrono`, `chrono-tz`	Time handling, default zone from the system
`tracing`, `tracing-appender`	Rolling local logs
`anyhow`, `thiserror`	Errors

Tauri plugins: `single-instance`, `autostart`, `notification`, `opener`. Do not use the Tauri `global-shortcut` plugin for the modifier-only chords.

Dev dependencies: `wiremock` (mock Gemini server for integration tests) and `tokio-test`.

3.3 Process and window model

One process. Three kinds of window, all created by Rust:

1. **Island** (`island`): transparent, frameless, always-on-top, no taskbar entry, top-centre, click-through while idle.
2. **Overlay** (`overlay`): full-screen transparent window per monitor, created only for Circle to Ask and for Computer Use cursor feedback, destroyed after use.
3. **Main** (`main`): normal window for settings, history, to-do, routines. Hidden until opened from the tray or island. Closing it hides it; Quit is in the tray menu.

3.4 Module map (Rust)

Module	Responsibility
`hotkeys`	Dedicated thread with the LL hook and message loop; emits `ChordDown/ChordUp` events over a channel
`audio`	Start/stop capture, return a PCM buffer; live level meter events

Module	Responsibility
`capture`	Monitor enumeration, screenshots, cropping to a drawn region
`gemini`	`client` (REST, retries, 429 backoff), `live` (WebSocket), `computer_use`, `prompts`, `schema` (tool declarations)
`tools`	Local tool implementations the model can call (section 4.4)
`input`	`SendInput` wrappers: keys, text, mouse, scroll, drag
`agent`	Orchestrator: turns an utterance into model calls and tool execution; owns the Computer Use loop and its safety gate
`scheduler`	Loads schedules from SQLite, fires jobs, handles missed runs
`store`	SQLite access, migrations, settings
`secrets`	Credential Manager access
`island`	Island state machine and events to the UI
`media`, `clipboard`, `wallpaper`, `apps`	Windows integrations used by the island panels and tools

3.5 Front end layout

`src/island`, `src/overlay`, `src/main` are three Vite entry points. Shared code in `src/shared` (typed IPC wrappers, Zustand stores, UI kit). The UI MUST NOT contain business logic beyond view state.

3.6 IPC contract

Tauri commands (front end to Rust) are listed in Appendix B. Events (Rust to front end) use the form `wisp://<topic>`; topics are in Appendix B. Every command returns `Result<T, UiError>` where `UiError` has `code` and a human-readable `message`; messages never contain secrets.

4. Functional requirements

Ten features, each with numbered requirements. Every feature reports its state to the island (idle, listening, thinking, acting, confirm, error) through the `island` module (section 4.9).

4.1 Setup and key storage (SET)

ID	Level	Requirement
FR-SET-01	MUST	A first-run wizard with four steps: (1) privacy notice, (2) get a key, (3) paste and validate the key, (4) microphone test
FR-SET-02	MUST	The privacy notice states that audio, screenshots and clipboard text sent for a request go to Google, and that on the free tier Google may use them to improve its products. The user must tick "I understand" to continue
FR-SET-03	MUST	Step 2 opens `https://aistudio.google.com/apikey` in the default browser. The app never automates that page
FR-SET-04	MUST	Step 3 validates the key by calling the Gemini models list endpoint. Success shows the count of models available; failure shows the HTTP status and a plain-language cause (invalid key, quota, no network)
FR-SET-05	MUST	The key is stored only in Credential Manager. The UI shows it masked (last 4 characters). Replace and Delete actions exist in Settings
FR-SET-06	MUST	Step 4 shows a live microphone level meter and a Test button that records 3 s and plays it back. If Windows blocks mic access, show a button that opens `ms-settings:privacy-microphone`

ID	Level	Requirement
FR-SET-07	MUST	Settings has a Models page: one dropdown per role (section 5.2) filled from the models list, with the defaults preselected and a warning marker on any default that is not in the list
FR-SET-08	SHOULD	Autostart at login (default on), tray icon with Open, Pause hotkeys, Quit. Single instance only: a second launch focuses the first
FR-SET-09	SHOULD	Local counters of requests today and HTTP 429 count; no prompt or audio content is ever stored in them

AC: on a fresh Windows 11 user profile, a user can finish the wizard in under 3 minutes and then dictate a sentence.

4.2 Global hotkeys (HK)

All chords are detected by one LL keyboard hook (section 7.1). They are Hold chords: active while down, ended on release.

Chord	Action	Notes
Hold Ctrl+Win	Dictation (4.3)	Either Ctrl and either Win key
Hold Ctrl+Shift	Circle to Ask (4.5)	Either Ctrl and either Shift key
Hold Ctrl+Alt+Space	Command mode (4.4)	Hold, speak, release
Esc pressed 3 times within 1.5 s	Emergency stop	Active whenever the agent is running; also the Stop button on the island

ID	Level	Requirement
FR-HK-01	MUST	A chord activates only after its keys have been held 120 ms with no other non-modifier key pressed. This avoids firing during normal Windows shortcuts such as Ctrl+Win+D

ID	Level	Requirement
FR-HK-02	MUST	If another non-modifier key is pressed before activation, the chord is abandoned and all keys pass through untouched
FR-HK-03	MUST	When a Win-key chord ends, the hook injects the unassigned virtual key `0xE8` down and up before the Win key release reaches the system, so the Start menu does not open. Injected events carry a private `dwExtraInfo` marker that the hook ignores
FR-HK-04	MUST	The hook callback does no work beyond updating key state and pushing an event to a channel. It MUST return in under 5 ms
FR-HK-05	MUST	If the hook is removed by Windows (callback timeout), a watchdog reinstalls it within 2 s and logs a warning
FR-HK-06	SHOULD	Chords are remappable in Settings from a fixed list of allowed modifier combinations, with conflict warnings
FR-HK-07	SHOULD	A tray toggle Pause hotkeys disables all chords without quitting

Known conflict: with several input languages installed, Windows may switch keyboard layout on Ctrl+Shift. The Settings page MUST show a note with the path Windows Settings, Time & language, Typing, Advanced keyboard settings, Input language hot keys, so the user can set it to Not Assigned.

4.3 Dictation (DIC)

ID	Level	Requirement
FR-DIC-01	MUST	While Ctrl+Win is held, the island shows Listening with a live level meter and audio is captured from the default input device
FR-DIC-02	MUST	Recordings under 300 ms are discarded silently. Recordings stop automatically at 120 s

ID	Level	Requirement
FR-DIC-03	MUST	On release, the audio (16 kHz mono 16-bit PCM, wrapped as WAV) is sent in one request to the dictation model with the cleanup instruction in 5.3
FR-DIC-04	MUST	Cleanup rules: remove filler words and false starts; when the speaker corrects themselves the last version wins; add punctuation and capitalization; keep the speaker's meaning and language; add nothing; return only the final text with no quotes or commentary
FR-DIC-05	MUST	If the result is empty or the model reports it could not understand, insert nothing and show Didn't catch that on the island for 2 s
FR-DIC-06	MUST	Insertion: save the clipboard text, place the result on the clipboard, send Ctrl+V to the focused window, then restore the old clipboard after 300 ms. If the clipboard holds non-text data, use FR-DIC-07 instead (see 7.5)
FR-DIC-07	MUST	If the paste fails or the target blocks it, fall back to typing the text with `SendInput` Unicode events. If the foreground window is elevated, show a notification explaining that Windows blocks input and copy the text to the clipboard instead
FR-DIC-08	SHOULD	A personal dictionary (up to 200 terms) is added to the prompt so names and jargon are spelled as the user wants
FR-DIC-09	SHOULD	The last 50 dictations are kept for 7 days in SQLite with a Copy button in the main window. A toggle turns history off and deletes it

AC: speaking "uh so meet at nine… no, ten?" inserts text that contains "10" or "ten" and not "nine". Release-to-insert time is at most 4 s at p90 for a 10 s clip on a 50 Mbps connection.

4.4 Voice commands (CMD)

ID	Level	Requirement
FR-CMD-01	MUST	While Ctrl+Alt+Space is held, the island shows Listening. On release, the audio is sent in one request to the command model together with the system prompt (5.3) and the function declarations of the enabled tools
FR-CMD-02	MUST	Agent loop: execute each function call the model returns with the matching local tool, send the results back as function responses, and repeat until the model replies with text or 8 rounds have run
FR-CMD-03	MUST	The final text appears on the island for up to 6 s. There is no spoken reply in v1
FR-CMD-04	MUST	Typed fallback: clicking the island opens a one-line text box that feeds the same pipeline
FR-CMD-05	MUST	App resolution: index Start Menu shortcuts (all-users and per-user Programs folders) and installed Store apps at startup and every 30 min. Match names by normalized fuzzy score. If the best score is clearly ahead, launch it; if two are close, show up to 3 chips on the island to pick
FR-CMD-06	MUST	Web search builds the URL for a named engine and opens it in the default browser. Default engine: Google. Supported by name: Google, YouTube, Amazon (amazon.in), Wikipedia, Bing, DuckDuckGo, GitHub, Maps. Only `http` and `https` URLs may be opened
FR-CMD-07	MUST	Only tools enabled in the active Loadout (4.10) are declared to the model
FR-CMD-08	MUST	A tool error is returned to the model as a function response. After 2 consecutive tool failures the loop stops and the island shows the error

ID	Level	Requirement
FR-CMD-09	SHOULD	The main window lists the last 100 commands: transcript, tools called with arguments, outcome, duration

Tool set for v1 (the `tools` module, declared in `gemini::schema`):

Tool	Arguments	Effect	Confirm
`open_app`	`name`	Launch an app resolved per FR-CMD-05	No
`open_url`	`url`	Open an http/https URL in the default browser	No
`web_search`	`query`, `engine?`	Open the search results page	No
`type_text`	`text`	Insert text into the focused window (same path as FR-DIC-06/07)	No
`press_keys`	`keys` (for example `ctrl+s`)	Send a key chord. Blocked: Win+L, Ctrl+Alt+Del, Alt+F4 on the Wisp window	No
`clipboard_read`, `clipboard_write`	`text?`	Read or set clipboard text	No
`media_control`	`action` (`play_pause`, `next`, `previous`)	Control the current media session through GSMTC	No
`set_volume`	`level` (0 to 100) or `mute`	Set the default output device volume	No
`create_todo`	`title`, `due?`	Add a to-do (4.8)	No

Tool	Arguments	Effect	Confirm
`create_schedule`	`command`, `when` (structured, see 5.5), `name?`	Create a scheduled task (4.6)	No, a confirmation card is shown on the island but does not block
`create_routine`	`name`, `command`, `when`	Create a recurring routine (4.6)	Same as above
`set_wallpaper`	`path`	Set the desktop wallpaper from a local image	No
`describe_screen`	`question`	Capture the screen and answer with the vision model, returning text	No
`run_computer_task`	`goal`	Start the Computer Use loop (4.7)	Yes, first use per app session

No shell or PowerShell tool exists in v1. Computer Use covers anything else. Declaration order follows the milestones: `create_schedule` and `create_routine` from M4, `run_computer_task` from M5, `create_todo` from M6.

AC: "Search how to cook spaghetti on YouTube" opens the YouTube results page for that query within 4 s. "Open Spotify" launches Spotify if installed.

4.5 Circle to Ask (CTA)

ID	Level	Requirement
FR-CTA-01	MUST	When Ctrl+Shift activates (FR-HK-01), capture the monitor under the cursor first, then show a full-screen overlay on it with a crosshair cursor and 20% dim. The overlay window MUST be excluded from capture with `SetWindowDisplayAffinity(WDA_EXCLUDEFROMCAPTURE)`

ID	Level	Requirement
FR-CTA-02	MUST	While the chord is held, left-mouse drags draw freehand strokes with a glow. Several strokes are allowed
FR-CTA-03	MUST	Audio is recorded for the whole hold as the spoken question
FR-CTA-04	MUST	On release, the overlay closes and one request goes to the vision model with: the audio, a context image (the full screenshot scaled to at most 1568 px on its long edge with the strokes drawn in red), and a detail crop (bounding box of all strokes plus 24 px padding, clamped to the monitor, at native resolution up to 1568 px)
FR-CTA-05	MUST	With no strokes, send only the context image. With no audio (under 300 ms), show the result card with a text box so the user can type the question
FR-CTA-06	MUST	The answer appears in a small always-on-top card near the drawn region (or centred if none), rendered as Markdown, with Copy, Follow-up and Close. Esc closes it
FR-CTA-07	SHOULD	Follow-up keeps the same images and conversation for up to 5 turns, by typed text or by holding Ctrl+Shift again over the card
FR-CTA-08	MUST	All geometry is in physical pixels and correct on multiple monitors with different scaling
FR-CTA-09	MUST	Images are encoded as JPEG quality 85 (context) and PNG (crop), and the total inline payload stays under 4 MB; if larger, reduce size until it fits

AC: on a second monitor at 150% scaling, circling a table cell and asking "what is the total?" returns an answer that refers to the circled content, and the crop matches the circled area within 24 px.

4.6 Scheduler and routines (SCH)

ID	Level	Requirement
FR-SCH-01	MUST	A job has a name, a natural-language command, and a schedule: once at a date and time, daily, weekdays, weekly on chosen days, monthly on a day, or an advanced cron expression. Time zone is the system zone
FR-SCH-02	MUST	Jobs are created by voice (`create_schedule`, `create_routine`) or from a form in the main window
FR-SCH-03	MUST	When a job fires, its command text runs through the same pipeline as voice commands (4.4) with no audio. The model receives the text and the job's allowed tools
FR-SCH-04	MUST	Jobs persist in SQLite. At startup the scheduler loads them, computes the next run of each, and sleeps until the earliest. It recomputes whenever a job changes
FR-SCH-05	MUST	Missed runs: if the app was not running at the due time and the due time is at most 15 min ago, run now. Otherwise record the run as missed and show a notification. The grace window is a setting
FR-SCH-06	MUST	Unattended safety: scheduled runs have `run_computer_task` disabled unless the job has Allow Computer Use ticked, which can only be set in the main window, never by voice. If an allowed run reaches a confirmation request it pauses and notifies; with no answer in 10 min it aborts
FR-SCH-07	MUST	Each run stores status (ok, failed, missed, aborted), a one-line summary from the model, and duration, and shows a Windows notification with the summary
FR-SCH-08	MUST	If the Windows session is locked at run time, defer the job until unlock within the grace window, then run or mark missed

ID	Level	Requirement
FR-SCH-09	MUST	One job runs at a time; others wait in a queue of at most 5
FR-SCH-10	SHOULD	The main window lists jobs with enable/disable, edit, Run now and delete, plus 30 days of run history

The app does not wake a sleeping PC. A job due while the PC sleeps follows FR-SCH-05 on wake.

AC: "Every morning at 9, open Unity and Spotify" creates a daily 09:00 routine with command "Open Unity and Spotify", visible in the list, and at 09:00 the apps open and a notification reports it. "Search Amazon for gift wraps on 10th November at 11:00" creates a one-time job for that date.

4.7 Computer Use agent (CU)

The agent operates the desktop with the Gemini Computer Use tool in its desktop environment: it looks at a screenshot, proposes one action, Wisp executes it with real input, and the loop repeats. The wire format MUST be taken from the official page in 5.1. Computer Use is a Preview capability, so this feature is designed to fail safe.

ID	Level	Requirement
FR-CU-01	MUST	A task starts from `run_computer_task`, or from a Run task box in the main window
FR-CU-02	MUST	First task per app session: show a confirmation card with the goal and a warning that Wisp will use mouse and keyboard. While a task runs the island shows a banner Wisp is controlling your PC, Esc x3 to stop
FR-CU-03	MUST	Loop: capture the target monitor, send goal plus history and the screenshot, receive the proposed action with its intent and any safety decision, validate, execute, capture again, return the result, repeat

ID	Level	Requirement
FR-CU-04	MUST	Target monitor: the primary monitor by default, selectable in Settings. Model coordinates (normalized 0 to 999) convert to pixels as `round(n / 1000 * size)` plus the monitor origin, in physical pixels
FR-CU-05	MUST	Execute the desktop-environment actions the docs list (click variants, mouse down/up, move, type, drag and drop, wait, key presses and hotkeys, scroll, screenshot) through the `input` module. An unknown action returns an error result to the model and does not crash the loop
FR-CU-06	MUST	Limits: 40 steps and 10 min per task, 60 s per model call. If three consecutive steps produce the same action and the same screenshot hash, abort as stuck
FR-CU-07	MUST	Safety decisions: `allowed` proceeds; `require_confirmation` pauses and shows the model's intent on a confirm card, then sends the acknowledgement only if the user taps Yes; `blocked` stops the task. Never auto-approve. A card unanswered after 5 min counts as No
FR-CU-08	MUST	Independent guard: if the foreground window's process or title matches the guard list, pause for confirmation whatever the model says. Default guard list (editable): `1Password.exe`, `Bitwarden.exe`, `KeePass.exe`, `KeePassXC.exe`, and titles containing bank, payment or checkout
FR-CU-09	MUST	Enable the model's prompt-injection detection option when the selected model supports it (the docs say 3.5 Flash-Lite or later)
FR-CU-10	MUST	Emergency stop (Esc x3 or Stop button) cancels the in-flight request, releases any held keys or mouse buttons with up events, closes overlays, and ends the task within 1 s

ID	Level	Requirement
FR-CU-11	MUST	Feedback: a click-through overlay shows a ring at each target point for 400 ms, and the island shows the model's intent text for each step
FR-CU-12	MUST	Audit log per step: time, intent, action and arguments, safety decision, user answer. Screenshots are held in memory only. A Debug setting, off by default, saves them to `%APPDATA%/Wisp/debug` and deletes them after 24 h
FR-CU-13	MUST	Cannot operate on elevated windows, the UAC secure desktop or the lock screen. If the foreground process is elevated or the screenshot is black, stop and explain why
FR-CU-14	SHOULD	User takeover: a physical (non-injected) key or mouse button press during a task pauses it; the island offers Resume and Stop
FR-CU-15	MUST	The result (final model text or failure reason) shows on the island and is saved to history

AC: the task "Open Notepad and type hello world" finishes in under 12 steps with the text present. Pressing Esc three times during any task ends it in under 1 s with no stuck keys.

4.8 To-do list (TODO)

ID	Level	Requirement
FR-TODO-01	MUST	A to-do has a title, optional notes, optional due date and time, a done flag, and created and completed timestamps
FR-TODO-02	MUST	Voice: "add call Mom to my to-do list" and "add gym at 6 PM" call `create_todo`; a spoken time becomes the due time
FR-TODO-03	MUST	At the due time the app shows a Windows notification with Done and Snooze 10 min. This uses the scheduler's timer, not the model

ID	Level	Requirement
FR-TODO-04	MUST	The main window lists to-dos with filters Today, Upcoming, Done, supports add, edit, tick, delete and drag to reorder
FR-TODO-05	SHOULD	Completed items are deleted after 30 days

AC: "add gym at 6 PM" creates an item due today at 18:00 (tomorrow if it is already past), and a notification appears at 18:00.

4.9 Island widget (ISL)

ID	Level	Requirement
FR-ISL-01	MUST	The island is a transparent, frameless, always-on-top window at top-centre of the chosen monitor, 8 px below the top edge. It never takes focus (`WS_EX_NOACTIVATE`), has no taskbar button, and is excluded from screen capture. Collapsed size is 140 x 34 logical px
FR-ISL-02	MUST	States and visuals: idle (small dot), listening (waveform from the live level), thinking (pulse), acting (spinner plus intent text), confirm (card with Yes and No), result (text for 6 s), error (red text for 4 s). Transitions take at most 200 ms
FR-ISL-03	MUST	Hovering for 400 ms or scrolling down on the pill expands it to a 420 x 300 logical px panel with tabs Music, Plans, Clipboard, Apps, Wallpapers, Quick settings. Scrolling cycles tabs. It collapses 800 ms after the pointer leaves
FR-ISL-04	MUST	While collapsed and idle, only the pill is clickable; the rest of the window ignores the mouse
FR-ISL-05	MUST	Music tab: current session title, artist, artwork, play/pause, next, previous and a progress bar from GSMTC, updated by event

ID	Level	Requirement
FR-ISL-06	MUST	Plans tab: today's to-dos with checkboxes and a count (for example 0/3), and the next 3 scheduled jobs
FR-ISL-07	SHOULD	Clipboard tab, off by default: last 20 text items captured with a clipboard format listener, kept in memory only and cleared on quit. Items flagged by the source app as excluded from clipboard history are skipped. Clicking an item pastes it into the previously focused window
FR-ISL-08	SHOULD	Apps tab: up to 12 pinned apps chosen from the app index; click launches
FR-ISL-09	SHOULD	Wallpapers tab: thumbnails of images in a folder the user picks; click sets the wallpaper
FR-ISL-10	SHOULD	Quick settings: pause hotkeys, choose microphone, mute, autostart
FR-ISL-11	SHOULD	When a full-screen app or presentation is in the foreground, hide the island while idle and show it when the assistant is active
FR-ISL-12	SHOULD	Respect the Windows setting that turns animations off

4.10 Customization slots (CUS)

Four slots make the assistant the user's own. Each slot has named presets saved as JSON and applies live, without restart.

Slot	What it controls	Built-in presets
Soul	Assistant name, tone, language preference, and an addendum of up to 500 characters appended to the command and answer prompts. It never changes the dictation cleanup rules	Default, Concise, Friendly

Slot	What it controls	Built-in presets
Loadout	Which tools from 4.4 are enabled	Minimal (open app, open URL, search), Standard (all except Computer Use), Full (all)
Skin	Island theme: light, dark or auto; accent colour; corner radius; opacity; font size	Midnight, Paper, Ember
Keys	Hotkey chords (FR-HK-06), default search engine, microphone, target monitor	Default

ID	Level	Requirement
FR-CUS-01	MUST	The default Loadout is Full
FR-CUS-02	MUST	Users can create, rename, duplicate and delete their own presets, and switch the active preset per slot from Settings or the island
FR-CUS-03	SHOULD	Export and import a preset as a `.json` file with a `schema_version` field; reject unknown versions with a clear message

5. Gemini API integration

All model calls go through the Rust `gemini` module over HTTPS with the user's key in the `x-goog-api-key` header. This section fixes which model does which job, the prompts, and the behaviour around errors. It deliberately does not restate wire formats, which change.

5.1 Sources the agent must read before coding each integration

Integration	Page
Computer Use (loop, actions, 0 to 999 coordinates, safety decisions)	Gemini Computer Use
Live API (optional path, 5.6)	Gemini Live API
Usage tiers and 429 behaviour	Rate limits

Integration	Page
Model IDs, free-tier availability, prices	Pricing and the models page in the same docs
Function calling, audio input, image input	The Function calling, Audio understanding and Image understanding pages in the same docs. Their URLs were not verified when this SRS was written: find them from the docs index

5.2 Model roles and defaults

Each role is a setting (FR-SET-07). Defaults below are the IDs named on the Computer Use page when this SRS was written; every default MUST be checked against the live models list and, if missing, the user is warned.

Role	Default model ID	Used by	Free tier
`dictation`	`gemini-3.5-flash-lite`	Dictation (4.3). Cheapest model that accepts audio	Yes
`command`	`gemini-3.8-flash`	Voice commands, scheduled jobs, schedule parsing (4.4, 4.6)	Yes
`vision`	`gemini-3.8-flash`	Circle to Ask, `describe_screen` (4.5)	Yes
`computer_use`	`gemini-3.8-flash` (the docs' recommendation). Lower-cost option: `gemini-3.5-flash-lite`	Computer Use (4.7)	Yes
`live` (optional)	Read the current Live model ID from the models list	Streaming path (5.6)	Yes

Pro and image-generation models need a paid plan and are not used.

5.3 Prompts

Prompts live in `src-tauri/src/gemini/prompts.rs` as constants and are the single place to tune behaviour.

Dictation cleanup (system instruction)

```
You turn spoken audio into clean written text.
Output ONLY the text the speaker meant to write. No quotes, no commentary.
Remove filler words and false starts. If the speaker corrects themselves,
keep only the corrected version. Add punctuation and capitalization.
Keep the speaker's wording, meaning and language. Do not add content.
Use these spellings when they occur: {dictionary}.
If the audio has no intelligible speech, output exactly: <EMPTY>
```

The app treats `<EMPTY>` or an empty string as FR-DIC-05.

Command mode (system instruction)

```
You are {assistant_name}, a voice assistant that controls a Windows 11 PC.
Tone: {tone}. {soul_addendum}
Now: {iso_datetime} ({timezone}). Resolve relative dates and times from this.
Use the provided tools to act. Never say an action succeeded unless a tool
result says so. Prefer one tool call per need. Keep replies under 20 words.
For tasks that need seeing or clicking the screen, call run_computer_task.
If the request is unclear and a wrong guess would be costly, ask one short
question.
```

Circle to Ask (system instruction)

```
The user circled part of their screen (red strokes in the first image; the
second image is a close crop). The audio is their question about it.
Answer concisely in Markdown, using what is visible. If the audio is unclear
or the crop does not contain what they ask about, say so briefly.
```

5.4 Request handling

ID	Level	Requirement
FR-GEM-01	MUST	Use the REST generate content endpoint with inline audio and image parts for dictation, commands and Circle to Ask. Dictation and Circle to Ask use no streaming

ID	Level	Requirement
FR-GEM-02	MUST	Timeouts: connect 5 s, total 30 s (60 s for Computer Use steps)
FR-GEM-03	MUST	Retry HTTP 429 and 5xx up to 3 times with delays of 1, 2 and 4 s plus jitter, honouring a `Retry-After` header when present. Never retry other 4xx
FR-GEM-04	MUST	Keep any single request under 10 MB; audio is at most 120 s of 16 kHz mono 16-bit PCM (about 3.8 MB) and images follow FR-CTA-09
FR-GEM-05	MUST	Map failures to these `UiError` codes: `NO_KEY`, `KEY_INVALID`, `RATE_LIMIT`, `NETWORK`, `MODEL_NOT_FOUND`, `SAFETY_BLOCKED`, `TIMEOUT`, `EMPTY_RESULT`, `INTERNAL`. Each has a one-line island message
FR-GEM-06	MUST	On `RATE_LIMIT` after retries, show Gemini rate limit reached, try again shortly, and increment the 429 counter (FR-SET-09)
FR-GEM-07	MUST	Audio, images, prompts and responses are never written to disk or logs, except the Debug mode of FR-CU-12

5.5 Structured schedule format

`create_schedule` and `create_routine` take a `when` object. The model fills it from the spoken phrase and the current date and time; Rust validates it and rejects past one-time dates.

```
{
  "kind": "once | daily | weekdays | weekly | monthly | cron",
  "date": "YYYY-MM-DD",
  "time": "HH:MM",
  "days": ["mon", "wed"],
  "day_of_month": 15,
  "cron": "0 9 * * 1-5"
}
```

Required fields by kind: `once` needs `date` and `time`; `daily` and `weekdays` need `time`; `weekly` needs `days` and `time`; `monthly` needs `day_of_month` and `time`; `cron` needs `cron`.

5.6 Live API (optional, after milestone M7)

v1 uses push-to-talk request and response, which is simpler and stays inside free-tier limits. If time allows after M7, a `live_mode` setting MAY switch Command mode to a streaming session: client to server WebSocket, 16 kHz 16-bit PCM in, audio output ignored, input transcription on, function calling through the same tool set, and ephemeral tokens are not needed because the key never leaves Rust.

5.7 Computer Use loop requirements

1. Follow the page in 5.1 for the request shape, the function call and result messages, and the `safety_decision` and acknowledgement fields.
2. Keep conversation history within the limits the page recommends; if it advises dropping old screenshots, do so.
3. Capture screenshots at the monitor's native resolution unless the page states a preferred size.
4. Treat every field in a model response as untrusted: validate coordinates are within 0 to 999, text length under 4,000 characters, and key names against an allowlist before executing.

6. Data model and storage

Everything is local. One SQLite database holds all user data; the API key is the only secret and lives in Credential Manager.

6.1 Locations

What	Where
Database	`%APPDATA%/Wisp/wisp.db`
Logs	`%LOCALAPPDATA%/Wisp/logs/` (daily rolling, 7 days kept)
Debug screenshots (off by default)	`%APPDATA%/Wisp/debug/` (deleted after 24 h)

What	Where
API key	Credential Manager, target `Wisp/GeminiApiKey`

Logs record events and error codes only: timestamps, module, tool name, status, duration. They never contain audio, screenshots, transcripts, prompts, clipboard text or the key.

6.2 Schema (version 1)

Migrations use `refinery`, files in `src-tauri/migrations/`. Timestamps are UTC ISO-8601 text. Booleans are 0 or 1.

```
CREATE TABLE settings (
  key        TEXT PRIMARY KEY,
  value_json TEXT NOT NULL
);

CREATE TABLE presets (
  id         INTEGER PRIMARY KEY,
  slot       TEXT NOT NULL CHECK (slot IN ('soul','loadout','skin','keys')),
  name       TEXT NOT NULL,
  json       TEXT NOT NULL,
  is_builtin INTEGER NOT NULL DEFAULT 0,
  is_active  INTEGER NOT NULL DEFAULT 0,
  UNIQUE (slot, name)
);

CREATE TABLE todos (
  id           INTEGER PRIMARY KEY,
  title        TEXT NOT NULL,
  notes        TEXT,
  due_at       TEXT,
  done         INTEGER NOT NULL DEFAULT 0,
  sort_order   INTEGER NOT NULL DEFAULT 0,
  created_at   TEXT NOT NULL,
  completed_at TEXT,
  snoozed_until TEXT
);

CREATE TABLE jobs (
  id               INTEGER PRIMARY KEY,
  name             TEXT NOT NULL,
```

```
  command          TEXT NOT NULL,
  when_json        TEXT NOT NULL,
  enabled          INTEGER NOT NULL DEFAULT 1,
  allow_computer_use INTEGER NOT NULL DEFAULT 0,
  next_run_at      TEXT,
  created_at       TEXT NOT NULL
);

CREATE TABLE job_runs (
  id          INTEGER PRIMARY KEY,
  job_id      INTEGER NOT NULL REFERENCES jobs(id) ON DELETE CASCADE,
  due_at      TEXT NOT NULL,
  started_at  TEXT,
  status      TEXT NOT NULL CHECK (status IN
('ok','failed','missed','aborted')),
  summary     TEXT,
  duration_ms INTEGER
);

CREATE TABLE command_history (
  id          INTEGER PRIMARY KEY,
  at          TEXT NOT NULL,
  source      TEXT NOT NULL CHECK (source IN
('voice','typed','job','agent_task')),
  transcript  TEXT,
  tools_json  TEXT,
  outcome     TEXT NOT NULL,
  duration_ms INTEGER
);

CREATE TABLE dictation_history (
  id   INTEGER PRIMARY KEY,
  at   TEXT NOT NULL,
  text TEXT NOT NULL
);

CREATE TABLE agent_steps (
  id           INTEGER PRIMARY KEY,
  history_id   INTEGER NOT NULL REFERENCES command_history(id) ON DELETE
CASCADE,
  at           TEXT NOT NULL,
  step         INTEGER NOT NULL,
  intent       TEXT,
```

```
  action_json  TEXT NOT NULL,
  safety       TEXT,
  user_answer  TEXT
);

CREATE TABLE pinned_apps (
  id        INTEGER PRIMARY KEY,
  app_id    TEXT NOT NULL UNIQUE,
  label     TEXT NOT NULL,
  position  INTEGER NOT NULL
);

CREATE TABLE usage_counters (
  day     TEXT NOT NULL,
  metric  TEXT NOT NULL,
  value   INTEGER NOT NULL DEFAULT 0,
  PRIMARY KEY (day, metric)
);
```

6.3 Retention

Data	Kept
`dictation_history`	50 rows and 7 days, whichever is smaller; off by a setting
`command_history` and `agent_steps`	100 commands, 90 days
`job_runs`	30 days
Completed `todos`	30 days
`usage_counters`	30 days

A cleanup task runs at startup and daily at 03:00 local time.

6.4 Settings keys

Settings are JSON values by key. Defaults are in Appendix A. Keys: `models.*` (one per role), `hotkeys.*`, `search.default_engine`, `audio.input_device`, `capture.target_monitor`, `island.monitor`, `island.hide_in_fullscreen`, `dictation.dictionary`, `dictation.history_enabled`, `clipboard.history_enabled`,

`scheduler.grace_minutes`, `agent.guard_list`, `agent.debug_screenshots`, `agent.max_steps`, `agent.max_minutes`, `app.autostart`, `app.first_run_done`, `app.privacy_ack_at`, `wallpaper.folder`.

7. Windows integration specifics

These are the parts most likely to go wrong. All are Win32 or WinRT calls made through the `windows` crate unless stated. Each needs a manual test checklist (rule 6 in 1.4).

7.1 Keyboard hook and chord state machine

- Install `WH_KEYBOARD_LL` with `SetWindowsHookExW` on one dedicated thread that runs a `GetMessageW` loop. The callback only updates a key-state set and sends an event on a channel (FR-HK-04).
- Ignore events with the `LLKHF_INJECTED` flag when they carry Wisp's `dwExtraInfo` marker (constant `0x57495350`). Treat other injected events (from other tools) as real.
- Never swallow modifier keys. Swallow only the Space key of Ctrl+Alt+Space while that chord is Active, so the focused app does not receive a space.
- Esc is never swallowed. The hook counts Esc presses for the emergency stop.
- Chord states: `Idle` to `Arming` when the chord's modifiers are all down; to `Active` after 120 ms still down with no other non-modifier key; back to `Idle` when any chord key is released. If a non-modifier key arrives during `Arming`, go to `Idle` and pass everything through.
- Win-key chords: before the Win key release is delivered, inject `0xE8` down and up (marker set) to prevent the Start menu (FR-HK-03).
- Watchdog (FR-HK-05): every 2 s compare `GetLastInputInfo` with the time of the last hook callback. If input happened more than 5 s after the last callback, uninstall and reinstall the hook.
- Windows removes a low-level hook whose callback exceeds the `LowLevelHooksTimeout` limit, so no allocation-heavy work or locks in the callback.

7.2 Audio capture

- Use `cpal` with the default input device (or the one chosen in Settings). Accept whatever format the device offers, convert to mono, resample to 16 kHz with `rubato`, store as 16-bit PCM in a pre-allocated buffer sized for 120 s.
- Emit the live level (RMS, 30 events/s) to the island.

- Handle device removal during capture by ending the recording and showing an error.
- Capture starts within 150 ms of chord activation. Keep the stream open but paused to avoid start latency only if measured start latency exceeds 150 ms.

7.3 Screen capture and DPI

- Declare the process Per-Monitor-V2 DPI aware in the application manifest, so every coordinate is in physical pixels.
- Enumerate monitors and capture with `xcap`. The monitor under the cursor (Circle to Ask) or the configured monitor (Computer Use) is used.
- Set `SetWindowDisplayAffinity(hwnd, WDA_EXCLUDEFROMCAPTURE)` on the island, overlay and result card windows so they never appear in captures.
- Content from protected video or DRM windows may capture black. Treat a fully black capture as a failure (FR-CU-13).

7.4 Input injection

Need	Method
Text	`SendInput` with `KEYEVENTF_UNICODE`, one UTF-16 unit per event, 5 ms between characters
Key or hotkey	Modifiers down in order, key down and up, modifiers up in reverse. Key names map through an allowlist table to virtual-key codes
Mouse move and click	Absolute coordinates with `MOUSEEVENTF_ABSOLUTE` and `MOUSEEVENTF_VIRTUALDESK`, normalized to 0 to 65535 across the virtual desktop
Scroll	`MOUSEEVENTF_WHEEL` in multiples of 120
Drag	Button down, 10 or more interpolated moves 10 ms apart, button up
Pacing	30 to 80 ms between Computer Use actions, so applications can react

The `input` module tracks every key and button it has pressed. On abort or panic it releases all of them. Because of UIPI, input into elevated windows is dropped by Windows; detect and report it (FR-CU-13, FR-DIC-07).

7.5 Clipboard

- Use the raw clipboard API (`OpenClipboard`, `GetClipboardData`, `SetClipboardData` with `CF_UNICODETEXT`). Retry opening up to 5 times, 20 ms apart, because other programs may hold it.
- Dictation paste (FR-DIC-06) saves and restores clipboard text only. If the clipboard currently holds any non-text format, skip the paste and use the typing path (FR-DIC-07), so the user's clipboard content is never lost.
- The clipboard history listener uses `AddClipboardFormatListener` on a hidden message window. Skip content flagged by the source with the formats `ExcludeClipboardContentFromMonitorProcessing` or `CanIncludeInClipboardHistory` set to 0.

7.6 App index and launch

- Index `.lnk` files under `%ProgramData%/Microsoft/Windows/Start Menu/Programs` and `%APPDATA%/Microsoft/Windows/Start Menu/Programs`. Store display name, target and shortcut path.
- Index Store apps by enumerating the `shell:AppsFolder` virtual folder through the shell COM interfaces. As an implementation fallback only, the same list may come from a hidden `powershell -NoProfile -Command Get-StartApps` call run by Wisp itself; the model can never trigger that.
- Launch shortcuts with `ShellExecuteW`. Launch Store apps with `explorer.exe shell:AppsFolder/<AppUserModelID>`.

7.7 Media, volume, wallpaper

- Media: `GlobalSystemMediaTransportControlsSessionManager`, current session, `TryTogglePlayPauseAsync`, `TrySkipNextAsync`, `TrySkipPreviousAsync`, and the `MediaPropertiesChanged` and `PlaybackInfoChanged` events.
- Volume: Core Audio `IAudioEndpointVolume` on the default render device.
- Wallpaper: `SystemParametersInfoW` with `SPI_SETDESKWALLPAPER` and `SPIF_UPDATEINIFILE | SPIF_SENDCHANGE`.

7.8 Session, power, full screen, elevation

- Lock and unlock: `WTSRegisterSessionNotification` for `WTS_SESSION_LOCK` and `WTS_SESSION_UNLOCK` (FR-SCH-08).

- Full-screen detection: `SHQueryUserNotificationState` (FR-ISL-11).
- Elevation of the foreground window: get its process ID, open the process with limited rights, read `TokenElevation`. If the process cannot be opened, assume it is elevated.
- Autostart: the Tauri autostart plugin (writes `HKCU` Run). No services, no drivers, no admin.

7.9 Antivirus note

The app combines a keyboard hook, input injection and screen capture and is unsigned, so Windows SmartScreen or Defender may flag it. This is accepted for a personal tool; the manual says to allow the app once and add an exclusion for its install folder if needed.

8. UI and UX specification

The look is a quiet dark-glass pill that stays out of the way until the assistant is working. All artwork, names and copy are original; nothing is copied from other products.

8.1 Windows

Window	Size	Behaviour
Island	140 x 34 collapsed, 420 x 300 expanded (logical px)	Section 4.9. Never takes focus
Overlay	Full monitor	Circle to Ask drawing and Computer Use target rings. Exists only while needed
Result card	360 wide, height to content, max 420	Circle to Ask answers and follow-up. Topmost, focusable, closes with Esc
Main	960 x 640, min 720 x 480	Opened from the tray or island. Close hides it
First-run wizard	560 x 520, centred	Runs in the main window until done

8.2 Main window pages

Page	Contents
Home	Today's to-dos, next jobs, last 5 commands, a Run task box (FR-CU-01), key status
To-do	Section 4.8
Routines	Section 4.6: list, form, run history
History	Commands with expandable steps (FR-CMD-09), dictation history (FR-DIC-09)
Customize	The four slots (4.10): preset picker, editor, import and export
Settings	General (autostart, island monitor), Models (FR-SET-07), Hotkeys (FR-HK-06 and the Windows Ctrl+Shift note), Audio, Privacy (history toggles, delete all data, debug mode), Safety (guard list FR-CU-08, grace minutes), About

A left rail lists the pages; the active page is highlighted. Delete all data wipes the database and logs after a confirmation, and keeps the key unless the user also ticks Remove key.

8.3 Visual system

- Font: Segoe UI Variable with Segoe UI as fallback; island text 13 px, main window body 14 px.
- Island: near-black glass (`rgba(16,16,20,0.82)`), 1 px inner border at 8% white, full-pill corners, accent colour only on the active state.
- Main window follows the Skin slot: light, dark or auto, one accent colour, 8 px corner radius, 8 px spacing grid.
- Icons: `lucide-react` . Motion: 120 to 200 ms ease-out; none when animations are disabled (FR-ISL-12).

8.4 Island messages

Situation	Text
Listening	A waveform, no text
Thinking	Thinking
Acting	The model's intent for the current step, truncated to 60 characters
Computer Use banner	Wisp is controlling your PC. Esc x3 to stop
`NO_KEY`	Add your Gemini key in Settings
`KEY_INVALID`	Gemini rejected the key
`RATE_LIMIT`	Gemini rate limit reached, try again shortly
`NETWORK`	No connection to Gemini
`MODEL_NOT_FOUND`	Model not available. Check Models in Settings
`SAFETY_BLOCKED`	Gemini declined this request
`TIMEOUT`	Gemini took too long
`EMPTY_RESULT`	Didn't catch that
`INTERNAL`	Something went wrong. Details are in the log

8.5 Confirmation card

Shown on the island (expanded) for Computer Use confirmations and the first Computer Use task. It shows the action in plain words, the model's stated intent, and two buttons, Yes and No. No is the default focus. Esc means No. It never auto-closes with Yes.

8.6 Accessibility

All main-window controls are reachable by keyboard with a visible focus ring, have text labels or `aria-label`, and meet 4.5:1 text contrast. Island state changes are exposed through an `aria-live` region. The result card is readable by a screen reader.

9. Non-functional requirements

Targets are measured on a typical Windows 11 laptop with a 50 Mbps connection. Network-dependent figures include Gemini's response time.

9.1 Performance

ID	Level	Requirement
NFR-01	MUST	Chord activation to recording start: 150 ms or less
NFR-02	MUST	Dictation release to text inserted: 4 s or less at p90 for clips up to 10 s
NFR-03	MUST	Circle to Ask release to answer visible: 6 s or less at p90
NFR-04	MUST	Hook callback duration: 5 ms or less (FR-HK-04)
NFR-05	MUST	Idle resource use with the island collapsed: under 1% CPU and under 200 MB RAM including WebView2
NFR-06	MUST	Cold start to hotkeys working: 3 s or less
NFR-07	SHOULD	Scheduled jobs start within 5 s of their due time
NFR-08	SHOULD	Emergency stop ends all activity within 1 s (FR-CU-10)

9.2 Security

ID	Level	Requirement
NFR-10	MUST	The API key is stored only in Credential Manager and read only by the Rust `secrets` module. It never crosses to the front end and never appears in logs

ID	Level	Requirement
NFR-11	MUST	The WebViews load only bundled local files. No remote content, no remote scripts. Content Security Policy: `default-src 'self'; img-src 'self' data:; style-src 'self' 'unsafe-inline'`
NFR-12	MUST	Tauri capabilities are least privilege: the front end may call only the commands in Appendix B. The shell and filesystem plugins are not enabled
NFR-13	MUST	Model output is data. It is never evaluated or executed. URLs are limited to http and https; key names, tool names and arguments are validated against allowlists and size limits before use
NFR-14	MUST	Network destinations are limited to Google's Gemini API host and URLs the user asked to open in their browser. No telemetry, analytics or update pings
NFR-15	MUST	Everything the screen shows to the model (Circle to Ask, Computer Use) is treated as possibly containing hostile instructions: the model's text is never trusted to change settings, tools, guard list or schedules except through the normal confirmed tool paths
NFR-16	MUST	`Allow Computer Use` on a job and edits to the guard list can only be changed from the main window with physical input, never through a tool call

9.3 Privacy

ID	Level	Requirement
NFR-20	MUST	First-run privacy notice per FR-SET-02, repeated in Settings, Privacy
NFR-21	MUST	Audio and images are held only in memory and discarded after the request (FR-GEM-07)

ID	Level	Requirement
NFR-22	MUST	Settings, Privacy offers Delete all data (8.2) and per-history toggles
NFR-23	SHOULD	Settings shows the current Gemini tier assumption and a note that enabling billing in Google AI Studio changes how Google may use the data

9.4 Reliability

ID	Level	Requirement
NFR-30	MUST	A tool failure or malformed model response never crashes the app; it becomes an error result
NFR-31	MUST	A panic hook and clean shutdown release all held keys and mouse buttons, stop recording and remove the hook
NFR-32	MUST	After resume from sleep, reinstall the hook and refresh the audio device
NFR-33	MUST	With no network, every feature that needs Gemini shows `NETWORK` within the timeout; local features (to-do, island panels, app launch by typed name through the island) keep working
NFR-34	MUST	Database writes use transactions. A corrupted database is renamed with a timestamp and a fresh one is created, with a notice

9.5 Compatibility

Windows 11 22H2 or later, x64. Evergreen WebView2. One to four monitors, scaling 100% to 200%. English UI.

9.6 Maintainability and quality gates

- Rust: `cargo fmt --check`, `cargo clippy -- -D warnings`, `cargo test` pass.
- Front end: `tsc --noEmit` with `strict`, ESLint and Vitest pass.

- Pure logic has unit tests: schedule parsing and next-run calculation, tool argument validation, coordinate conversion, app-name matching, chord state machine (with a fake event feed), prompt assembly. Target at least 70% line coverage on these modules.
- Commit `Cargo.lock` and `package-lock.json`. Pin major versions of Tauri, React and Vite.
- Settings, About has a Copy diagnostics button: app version, Windows build, model IDs, counters; no content or key.

10. Build, test and delivery

The app can only run on Windows 11. A coding agent working on Linux writes code and tests that compile on Windows through CI, and the user runs the built installer and the manual checklists on their PC.

10.1 User's PC setup (one time)

Tool	Requirement
Visual Studio Build Tools 2022	Workload Desktop development with C++, including the Windows 11 SDK
Rust	`rustup` with the `stable-x86_64-pc-windows-msvc` toolchain
Node.js	Current LTS (20 or newer) with npm
Git	Any recent version
WebView2 Runtime	Already present on Windows 11

Daily commands: `npm install`, `npm run tauri dev` (run with hot reload), `npm run tauri build` (installer).

10.2 Repository layout

```
wisp/
  package.json  vite.config.ts  tsconfig.json  tailwind.config.ts
  src/                        front end
    island/  overlay/  main/  shared/
```

```
src-tauri/
  Cargo.toml  tauri.conf.json  build.rs  capabilities/
  manifest.xml            Per-Monitor-V2 DPI manifest
  migrations/             SQL, refinery
  src/
    main.rs  lib.rs
    hotkeys/  audio/  capture/  input/  agent/  scheduler/
    store/  secrets/  island/  media/  clipboard/  wallpaper/  apps/
    gemini/{client,live,computer_use,prompts,schema}.rs
    tools/
  tests/                  integration tests with a mock Gemini server
docs/
  SRS.md                  this document, exported
  manual-tests/           one checklist per milestone
.github/workflows/build.yml
README.md  CHANGELOG.md
```

10.3 Continuous integration

Workflow `build.yml`, trigger on push and pull request, runner `windows-latest`:

1. Check out, set up Node (npm cache), install the Rust stable toolchain with `rust-cache`.
2. `npm ci`, then `npm run lint`, `npm run typecheck`, `npm test`.
3. `cargo fmt --check`, `cargo clippy -- -D warnings`, `cargo test`.
4. Build with `tauri-apps/tauri-action` for target `nsis` only, no signing, no release publishing.
5. Upload the installer as a workflow artifact kept 14 days.

10.4 Test strategy

Level	What	Where
Unit (Rust)	Pure logic listed in 9.6	`cargo test`
Unit (TS)	Stores, formatters, IPC wrappers (mocked)	Vitest

Level	What	Where
Integration	The `gemini` client against a local mock server: success, 429 with `Retry-After`, 5xx, malformed JSON, safety block, timeout. Tool loop with scripted function calls	`src-tauri/tests`
Manual (Windows)	Everything that touches hooks, audio, capture, input, overlay, notifications, media, wallpaper	`docs/manual-tests/M<n>.md`: numbered steps, expected result, a pass or fail box

The agent MUST write the manual checklist for a milestone before declaring it done, and the user ticks it on their PC.

10.5 Packaging

- Bundle target: `nsis` only. Install mode: per-user (`currentUser`), so no admin prompt.
- Product name: Wisp (working title). Identifier: `com.personal.wisp`. Version starts at `0.1.0`.
- No code signing, no auto-updater, no installer analytics. Updating means installing the next build over the old one; the database in `%APPDATA%/Wisp` is kept.

10.6 Working agreement for the agent

- One commit per requirement group, message format `<area>: <summary> (FR-XXX-nn)`.
- Keep `README.md` current: setup, run, build, where logs and data live.
- If a requirement cannot be met as written (for example the docs show a changed API shape), implement the closest safe behaviour, record the deviation in `docs/DEVIATIONS.md` with the requirement ID, and continue.

11. Milestones in build order

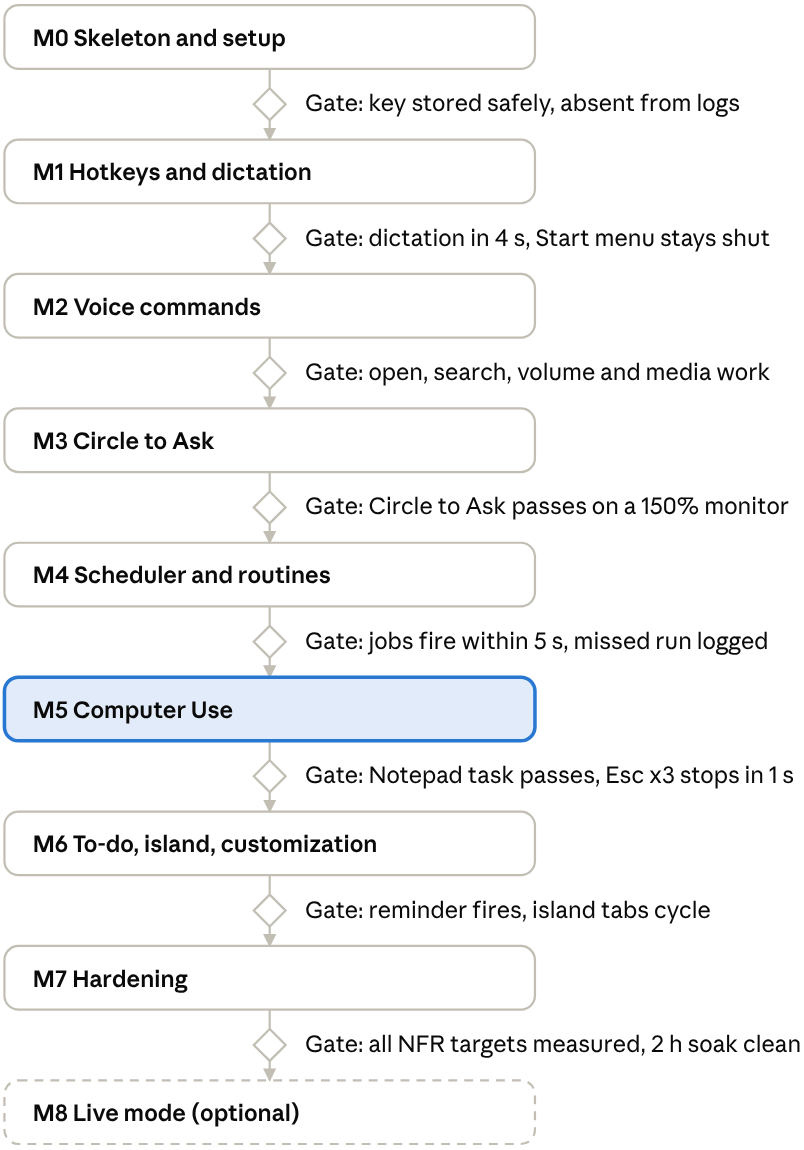
Nine milestones, built strictly in order. A milestone is done only when its code is merged, CI is green, its manual checklist exists, and every acceptance criterion in the table passes on the user's Windows 11 PC.

Milestone	Scope	Acceptance criteria
M0 Skeleton and setup	Tauri app, three windows stubbed, tray, single instance, autostart, logging, SQLite and migrations, settings store, Credential Manager, first-run wizard, CI producing an installer. FR-SET-01 to 09, NFR-10 to 12	Installer from CI installs without admin. The wizard validates a real key, stores it, and the key is absent from files and logs. Second launch focuses the first instance
M1 Hotkeys and dictation	LL hook and state machine, audio capture, Gemini client with retries and error codes, dictation, clipboard-safe insertion, minimal island states, dictation history. FR-HK-01 to 05, FR-DIC-01 to 09, FR-GEM-01 to 07	Holding Ctrl+Win in Notepad and speaking inserts clean text in 4 s or less; the Start menu never opens; unplugging the mic mid-recording shows an error and releases everything
M2 Voice commands	Command mode, tool loop, app index, search engines, key and clipboard tools, volume, media, wallpaper, typed fallback, Loadout filtering, command history. FR-CMD-01 to 09	"Open Spotify", "search how to cook spaghetti on YouTube", "set volume to 30", "pause the music" all work. A failing tool shows an error and the loop stops after 2 failures
M3 Circle to Ask	Overlay, monitor capture, strokes, crop, vision request, result card, follow-up. FR-CTA-01 to 09	The Circle to Ask AC in 4.5 passes on a 150% second monitor. The overlay never appears in the captured image
M4 Scheduler and routines	Job model, schedule parsing tool, scheduler loop, missed-run and lock handling, notifications, jobs UI. FR-SCH-01 to 10	The two example phrases in 4.6 create the right jobs; a job fires within 5 s of due time; closing the app for 20 min and reopening records a missed run

Milestone	Scope	Acceptance criteria
M5 Computer Use	Loop, coordinate mapping, action executor, safety decisions, guard list, kill switch, takeover pause, audit log, feedback rings. FR-CU-01 to 15	The Notepad task in 4.7 passes. Esc x3 stops any task in 1 s with no stuck keys. A page asking to enter a password triggers a confirmation card
M6 To-do, island, customization	To-do list and reminders, island panels, the four slots with presets and import/export. FR-TODO-01 to 05, FR-ISL-01 to 12, FR-CUS-01 to 03. The `create_todo` tool is declared from this milestone on	"Add gym at 6 PM" creates the item and a notification fires at 18:00. Scroll on the island cycles all six tabs. Changing a Skin preset applies without restart
M7 Hardening	Check every NFR, delete-all-data, diagnostics, sleep/resume handling, database recovery, final docs. NFR-01 to 34	All NFR targets measured and recorded in `docs/PERFORMANCE.md`; a 2-hour soak with hotkeys used every few minutes shows no hook loss and no memory growth over 50 MB
M8 Live mode (optional)	`live_mode` streaming command path (5.6)	Same tool results as the request path; first action starts at least 1 s sooner on a typical command

The order of the milestones and what gates each one is drawn below.

Each milestone must pass its gate before the next one starts



Highlighted: the riskiest milestone. Dashed: optional.

build order · 9 milestones, 8 gates

M0 to M7 are required. M8 is optional and starts only after M7 passes.

12. Risks, open questions and appendices

12.1 Risks

Risk	Impact	Mitigation
Ctrl+Win or Ctrl+Shift collide with Windows behaviour (Start menu, layout switch)	Hotkeys misfire	FR-HK-01 to 03, the Windows setting note in 4.2, test early in M1
Computer Use uses one request per step, so a 40-step task can hit free-tier per-minute or per-day limits	Tasks fail midway	Step cap, backoff (FR-GEM-03), cheaper model option (5.2), clear `RATE_LIMIT` message; the user can enable billing for higher limits
Computer Use is a Preview capability and may change	Breakage after Google updates	Isolated in `gemini::computer_use`, wire format read from docs at build time, deviations logged
Model IDs or free-tier access change	Calls fail with `MODEL_NOT_FOUND`	IDs are settings, verified against the models list (FR-SET-07)
Hostile text on screen or in a web page steers the agent (prompt injection)	Unwanted actions	Confirmation never auto-approved (FR-CU-07), guard list (FR-CU-08), injection detection (FR-CU-09), NFR-15, NFR-16
Free-tier data may be used by Google to improve products	Privacy	Notice in FR-SET-02, NFR-23; billing is a one-switch fix
Antivirus or SmartScreen flags the unsigned app	Install friction	7.9 note; per-user install
Elevated windows ignore injected input	Some apps cannot be controlled	Detect and explain (FR-CU-13, FR-DIC-07); no workaround
Protected video windows capture as black	Vision or agent cannot see them	Detect black capture and stop

Risk	Impact	Mitigation
Gemini prices for several models double on 1 January 2027	Higher cost on a paid plan only	None needed on the free tier; recheck pricing if billing is enabled

12.2 Open questions and the defaults assumed

Question	Default used in this SRS
Product name	Wisp, a working title, easy to rename in `tauri.conf.json` and the UI strings
Which monitor Computer Use controls	Primary monitor, selectable in Settings
Default search engine	Google
Speech language	English only
Keep dictation history by default	Yes, 7 days, toggle in Settings
Spoken replies from the assistant	Not in v1
Free key or billing	Start on free; switch later without code changes

Appendix A. Default settings

```
{
  "schema_version": 1,
  "models": {
    "dictation": "gemini-3.5-flash-lite",
    "command": "gemini-3.8-flash",
    "vision": "gemini-3.8-flash",
    "computer_use": "gemini-3.8-flash",
    "live": null
  },
  "hotkeys": {
    "dictation": "ctrl+win",
    "circle": "ctrl+shift",
    "command": "ctrl+alt+space",
    "emergency_stop": "esc*3"
  },
```

```
  "search": { "default_engine": "google" },
  "audio": { "input_device": null },
  "capture": { "target_monitor": "primary" },
  "island": { "monitor": "primary", "hide_in_fullscreen": true },
  "dictation": { "dictionary": [], "history_enabled": true },
  "clipboard": { "history_enabled": false },
  "scheduler": { "grace_minutes": 15 },
  "agent": {
    "guard_list": {
      "processes": ["1Password.exe", "Bitwarden.exe", "KeePass.exe",
"KeePassXC.exe"],
      "title_keywords": ["bank", "payment", "checkout"]
    },
    "debug_screenshots": false,
    "max_steps": 40,
    "max_minutes": 10
  },
  "app": { "autostart": true, "first_run_done": false, "privacy_ack_at": null
},
  "wallpaper": { "folder": null }
}
```

Appendix B. IPC contract

Commands (front end to Rust). Each returns `Result<T, UiError>`. Commands marked **main only** are rejected unless the calling window label is `main`, and no model tool path can reach them (NFR-16).

Area	Commands
Key and models	`secrets_status`, `secrets_set_key`, `secrets_delete_key`, `validate_key`, `list_models`
Settings	`settings_get`, `settings_set` (allowlisted keys only; `agent.guard_list` is **main only**)
Presets	`presets_list`, `presets_save`, `presets_activate`, `presets_delete`, `presets_export`, `presets_import`

Area	Commands
To-do	`todos_list`, `todos_create`, `todos_update`, `todos_delete`, `todos_reorder`, `todo_snooze`
Jobs	`jobs_list`, `jobs_create`, `jobs_update` (setting `allow_computer_use` is **main only**), `jobs_delete`, `jobs_run_now`, `job_runs_list`
Agent	`run_text_command`, `run_agent_task`, `agent_stop`, `agent_resume`, `confirm_answer` (id, yes)
History	`history_commands`, `history_dictations`, `copy_dictation`
Apps and media	`apps_search`, `apps_pin`, `apps_unpin`, `launch_app`, `media_control`, `wallpaper_list`, `wallpaper_set`, `clipboard_history_list`, `clipboard_history_paste`
System	`mic_devices_list`, `mic_test`, `monitors_list`, `window_show_main` (page), `delete_all_data` (remove_key) **main only**, `diagnostics_copy`

Events (Rust to front end), all prefixed `wisp://`:

Event	Payload
`island-state`	`state` (idle, listening, thinking, acting, confirm, result, error), `text?`, `code?`
`audio-level`	`rms` (0 to 1), about 30 per second while recording
`confirm-request`	`id`, `title`, `detail`, `kind` (first_task, safety, guard)
`agent-step`	`step`, `intent`, `x?`, `y?` (screen pixels for the target ring)
`overlay-init`	`monitor`, `width`, `height`, `scale`
`settings-changed`	`keys`

Event	Payload
`todos-changed`, `jobs-changed`, `media-changed`, `clipboard-changed`, `toast`	Small payloads, defined where each is first used